

Questions:

1. Turner syndrome has one X chromosome, choose the incorrect statement:

- ☐ a. Turner has a mild phenotype and patients are independent and can take care of themselves
- ☐ b. Turner cases are mostly infertile
- ☐ c. Monosomy X is of low effect, thus majority of conceptions continue to birth
- ☐ d. Turner syndrome is the only full monosomy that is seen in humans
- ☐ e. What is seen of Turner cases is only the tip of the iceberg, most cases are aborted

2. Choose the correct statement:

- ☐ a. Multiple aneuploidies happen in theory but are never seen in humans
- ☐ b. Sequential nondisjunction happens in both male and female gametogenesis simultaneously
- ☐ c. Complete nondisjunction will lead to trisomy
- ☐ d. Simultaneous parental nondisjunction happens in meiosis I and II in the same parent
- ☐ e. None is correct

3. Choose the incorrect statement about imprinting:

- ☐ a. Imprinting is stable in cells, but can differ according to tissue type or developmental stage
- ☐ b. Imprinting needs to be re-programmed with each generation
- ☐ c. None of the choices
- ☐ d. Imprinting depends on the parent of origin of the chromosome
- ☐ e. Imprinting can be exposed by uniparental disomy

4. Which of the following segregations could never result in a live birth?

- ☐ a. Interchange trisomy
- ☐ b. Tertiary trisomy
- ☐ c. Interchange monosomy
- ☐ d. Tertiary monosomy
- ☐ e. All have been reported in live birth babies

5. Choose the correct statement:

- a. Most chromosomal abnormalities can be seen at birth stage, earlier than that are not as frequent
 - b. Chromosomal abnormalities, when checked in different intervals of the life of human, have the same frequency
 - c. Most chromosomal abnormalities can be seen in older adults then live birth babies then blastocysts
 - d. Aging individuals are expected to have the most of constitutional chromosomal abnormalities in their cells than other life stages
 - e. Most chromosomal abnormalities can be seen in blastocysts then live birth babies then older adults
6. **One of the pathogenic mechanisms that arise from chromosomal abnormalities is position effect, how?**
- a. Can make a silent gene become active
 - b. Can convert an active gene into a silent one
 - c. Can cause a decrease in gene expression
 - d. Can make a gene increase its expression
 - e. All choices are right
7. **Choose the correct statement:**
- a. None is correct
 - b. Multiple aneuploidies happen in theory but never seen in humans
 - c. Complete nondisjunction will lead to trisomy
 - d. Simultaneous parental nondisjunction happens in meiosis I and II in the same parent
 - e. Sequential nondisjunction happens in both male and female gametogenesis simultaneously
8. **A child patient was found to have an autosomal recessive disease. Mutation testing indicated homozygosity for the pathogenic mutation. Mutation testing in the father found him heterozygous while the mother is wild type for both alleles. How could this happen?**
- a. The child inherited a chromosome from his paternal grandfather and the other homolog from his paternal grandmother
 - b. The child might be trisomic, then trisomic rescue happened and resulted in paternal isodisomy
 - c. The child might be trisomic, then trisomic rescue happened and resulted in maternal heterodisomy

- d. The child might be trisomic, then trisomic rescue happened and resulted in maternal isodisomy
- e. The child might be trisomic, then trisomic rescue happened and resulted in paternal heterodisomy

9. Choose the correct statement:

- a. Aging individuals are expected to have the most of constitutional chromosomal abnormalities in their cells than other life stages
- b. Most chromosomal abnormalities can be seen at birth stage, earlier than that are not as frequent
- c. Most chromosomal abnormalities can be seen in older adults then live birth babies then blastocysts
- d. Chromosomal abnormalities, when checked in different intervals of the life of human, have the same frequency
- e. Most chromosomal abnormalities can be seen in blastocysts then live birth babies then older adults

10. The most frequent autosomal reciprocal translocation in humans is between chr:

- a. 9 and 22
- b. X and Y
- c. 11 and 22
- d. 4 and 8
- e. 14 and 21

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	E	C	C	E	E	D	B	E	A

Questions (Continued):

11. Choose the incorrect statement about imprinting:

- a. Imprinting is stable in cells, but can differ according to tissue type or developmental stage
- b. Imprinting needs to be re-programmed with each generation

- c. None of the choices
- d. Imprinting depends on the parent of origin of the chromosome
- e. Imprinting can be exposed by uniparental disomy

12. Choose the correct statement:

- a. None is correct
- b. Multiple aneuploidies happen in theory but never seen in humans
- c. Complete nondisjunction will lead to trisomy
- d. Simultaneous parental nondisjunction happens in meiosis I and II in the same parent
- e. Sequential nondisjunction happens in both male and female gametogenesis simultaneously

13. Choose the correct statement:

- a. Aging individuals are expected to have the most of constitutional chromosomal abnormalities in their cells than other life stages
- b. Most chromosomal abnormalities can be seen at birth stage, earlier than that are not as frequent
- c. Most chromosomal abnormalities can be seen in older adults then live birth babies then blastocysts
- d. Chromosomal abnormalities, when checked in different intervals of the life of human, have the same frequency
- e. Most chromosomal abnormalities can be seen in blastocysts then live birth babies then older adults

14. Choose the correct statement:

- a. None is correct
- b. Multiple aneuploidies happen in theory but never seen in humans
- c. Complete nondisjunction will lead to trisomy
- d. Simultaneous parental nondisjunction happens in meiosis I and II in the same parent
- e. Sequential nondisjunction happens in both male and female gametogenesis simultaneously

15. Choose the correct statement:

- a. Aging individuals are expected to have the most of constitutional chromosomal abnormalities in their cells than other life stages

- b. Most chromosomal abnormalities can be seen at birth stage, earlier than that are not as frequent
- c. Most chromosomal abnormalities can be seen in older adults then live birth babies then blastocysts
- d. Chromosomal abnormalities, when checked in different intervals of the life of human, have the same frequency
- e. Most chromosomal abnormalities can be seen in blastocysts then live birth babies then older adults

Answers for Questions 11-15:

Q11	Q12	Q13	Q14	Q15
C	D	E	D	E